

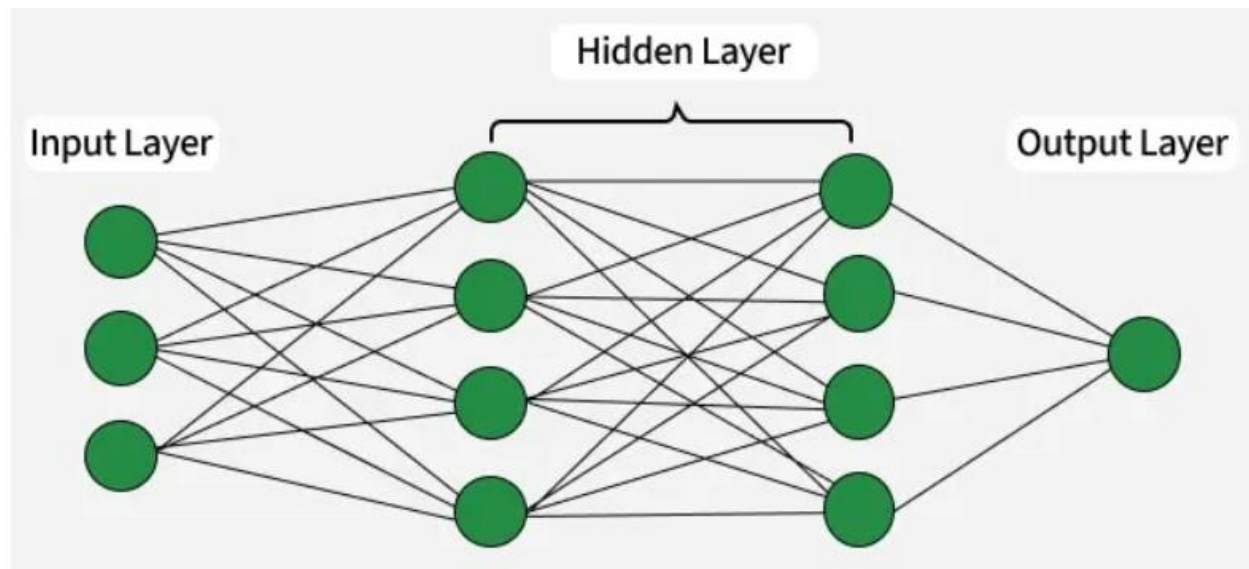
Introduction to Deep Learning

Deep Learning is transforming the way machines understand, learn and interact with complex data. Deep learning mimics neural networks of the human brain, it enables computers to autonomously uncover patterns and make informed decisions from vast amounts of unstructured data.

How Deep Learning Works?

Neural network consists of layers of interconnected nodes or neurons that collaborate to process input data. In a fully connected deep neural network data flows through multiple layers where each neuron performs nonlinear transformations, allowing the model to learn intricate representations of the data.

In a deep neural network the input layer receives data which passes through hidden layers that transform the data using nonlinear functions. The final output layer generates the model's prediction.



Difference between Machine Learning and Deep Learning

Aspect	Machine Learning	Deep Learning
Basic Idea	Applies statistical algorithms to learn patterns from data	Uses artificial neural networks to learn patterns from data
Data Requirement	Works well with small to medium datasets	Requires a large amount of data
Task Complexity	Better for simple and low-label tasks	Better for complex tasks like image and text processing
Training Time	Takes less time to train	Takes more time to train
Feature Extraction	Features are manually selected and extracted	Features are automatically extracted
Learning Process	Not end-to-end	End-to-end learning
Model Complexity	Less complex	Highly complex
Interpretability	Easy to understand and explain	Hard to interpret (black box)
Hardware Requirement	Can run on CPU, needs less computing power	Needs GPU and high-performance systems
Use Cases	Spam detection, recommendation systems	Image recognition, NLP, speech recognition

Types of neural networks

1. **Feedforward neural networks (FNNs)**: They are the simplest type of ANN, where data flows in one direction from input to output. It is used for basic tasks like classification.
2. **Convolutional Neural Networks (CNNs)**: They are specialized for processing grid-like data, such as images. CNNs use convolutional layers to detect spatial hierarchies, making them ideal for computer vision tasks.
3. **Recurrent Neural Networks (RNNs)**: They are able to process sequential data, such as time series and natural language. RNNs have loops to retain information over time, enabling applications like language modeling and speech recognition. Variants like LSTMs and GRUs address vanishing gradient issues.
4. **Generative Adversarial Networks (GANs)**: This consists of two networks—a generator and a discriminator—that compete to create realistic data. GANs are widely used for image generation, style transfer and data augmentation.
5. **Autoencoders**: They are unsupervised networks that learn efficient data encodings. They compress input data into a latent representation and reconstruct it, useful for dimensionality reduction and anomaly detection.
6. **Transformer Networks**: It has revolutionized NLP with self-attention mechanisms. Transformers excel at tasks like translation, text generation and sentiment analysis, powering models like GPT and BERT.